# THE SEPARATION AXIOMS FOR A MORPHISM OF FORMAL ALGEBRAIC SPACES CAN BE CHECKED ON AFFINE BASE CHANGES

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof displays five commutative diagrams and formulas of fibre products; all lie inside the counted span and are reproduced verbatim.

*Editorial note.* The counted spans carry 7 navigation stubs; no bibliography citation occurs in the spans.

*Editorial note.* The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

*Allowed standard prerequisites (editorial).*

- Stability of the separation axioms under base change. Same chapter, Lemma lemma-base-change-separated, invoked at proof lines 6913-6915. Gives the two immediate implications.
- The definition of the separation axioms for a morphism and of being of the corresponding type as a formal algebraic space. Same chapter, Definition definition-separated-morphism at statement lines 6887-6888, and Definition definition-separated at statement lines 6899-6900. Fix the properties whose equivalence is asserted.
- Affine formal algebraic spaces as filtered colimits, with factorisation of morphisms into the colimit through a member. Same chapter, Definition definition-affine-formal-algebraic-space at proof lines 6956-6958 and Lemma lemma-factor-through-thickening at proof lines 6970-6972. Underlies the passage from the third to the fifth formulation.
- The definition of a formal algebraic space through a presentation, and descent of representable transformation properties. Same chapter, Definition definition-formal-algebraic-space at proof lines 6917-6918 and 6974-6976, and Spaces, Lemma spaces-lemma-descent-representable-transformations-property at proof lines 6976-6981. Supply the presentation and transfer the property along the covering.

*Source reference.* Original definition 0AJ7: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 601–601. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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*Date*: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 0ARP: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 703–703. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original lemma 0AIA: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2080–2080. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original lemma 03KD: sources/stacks-chow/tags.txt, source lines 4673–4673. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

**Definition 0.1.** Let $S$ be a scheme. We say a sheaf $X$ on $(Sch/S)_{fppf}$ is an *affine formal algebraic space* if there exist

(1) a directed set $\Lambda$,
(2) a system $(X_\lambda, f_{\lambda\mu})$ over $\Lambda$ in $(Sch/S)_{fppf}$ where
   (a) each $X_\lambda$ is affine,
   (b) each $f_{\lambda\mu} : X_\lambda \to X_\mu$ is a thickening,

such that

$$X \cong \operatorname{colim}_{\lambda\in\Lambda} X_\lambda$$

as fppf sheaves and $X$ satisfies a set theoretic condition (see Remark 0AIS). A *morphism of affine formal algebraic spaces* over $S$ is a map of sheaves.

**Definition 0.2.** Let $S$ be a scheme. We say a sheaf $X$ on $(Sch/S)_{fppf}$ is a *formal algebraic space* if there exist a family of maps $\{X_i \to X\}_{i\in I}$ of sheaves such that

(1) $X_i$ is an affine formal algebraic space,
(2) $X_i \to X$ is representable by algebraic spaces and étale,
(3) $\coprod X_i \to X$ is surjective as a map of sheaves

and $X$ satisfies a set theoretic condition (see Remark 0AIS). A *morphism of formal algebraic spaces* over $S$ is a map of sheaves.

**Definition 0.3.** Let $S$ be a scheme. Let $f : X \to Y$ be a morphism of formal algebraic spaces over $S$. Let $\Delta_{X/Y} : X \to X \times_Y X$ be the diagonal morphism.

(1) We say $f$ is *separated* if $\Delta_{X/Y}$ is a closed immersion.
(2) We say $f$ is *quasi-separated* if $\Delta_{X/Y}$ is quasi-compact.

**Lemma 0.4.** *Let $S$ be a scheme. Let $f : X \to Y$ be a morphism of formal algebraic spaces over $S$. Let $\mathcal{P}$ be any of the separation axioms of Definition 0.3. The following are equivalent*

(1) *$f$ is $\mathcal{P}$,*
(2) *for every scheme $Z$ and morphism $Z \to Y$ the base change $Z \times_Y X \to Z$ of $f$ is $\mathcal{P}$,*

(3) *for every affine scheme $Z$ and every morphism $Z \to Y$ the base change $Z \times_Y X \to Z$ of $f$ is $\mathcal{P}$,*
(4) *for every affine scheme $Z$ and every morphism $Z \to Y$ the formal algebraic space $Z \times_Y X$ is $\mathcal{P}$ (see Definition 0AJ7),*
(5) *there exists a covering $\{Y_j \to Y\}$ as in Definition 0.2 such that the base change $Y_j \times_Y X \to Y_j$ has $\mathcal{P}$ for all $j$.*

*Proof.* We will repeatedly use Lemma 0ARP without further mention. In particular, it is clear that (1) implies (2) and (2) implies (3).

Assume (3) and let $Z \to Y$ be a morphism where $Z$ is an affine scheme. Let $U$, $V$ be affine schemes and let $a : U \to Z \times_Y X$ and $b : V \to Z \times_Y X$ be morphisms. Then

$$U \times_{Z \times_Y X} V = (Z \times_Y X) \times_{\Delta, (Z \times_Y X) \times_Z (Z \times_Y X)} (U \times_Z V)$$

and we see that this is quasi-compact if $\mathcal{P}$ ="quasi-separated" or an affine scheme equipped with a closed immersion into $U \times_Z V$ if $\mathcal{P}$ ="separated". Thus (4) holds.

Assume (4) and let $Z \to Y$ be a morphism where $Z$ is an affine scheme. Let $U$, $V$ be affine schemes and let $a : U \to Z \times_Y X$ and $b : V \to Z \times_Y X$ be morphisms. Reading the argument above backwards, we see that $U \times_{Z \times_Y X} V \to U \times_Z V$ is quasi-compact if $\mathcal{P}$ ="quasi-separated" or a closed immersion if $\mathcal{P}$ ="separated". Since we can choose $U$ and $V$ as above such that $U$ varies through an étale covering of $Z \times_Y X$, we find that the corresponding morphisms

$$U \times_Z V \to (Z \times_Y X) \times_Z (Z \times_Y X)$$

form an étale covering by affines. Hence we conclude that $\Delta : (Z \times_Y X) \to (Z \times_Y X) \times_Z (Z \times_Y X)$ is quasi-compact, resp. a closed immersion. Thus (3) holds.

Let us prove that (3) implies (5). Assume (3) and let $\{Y_j \to Y\}$ be as in Definition 0.2. We have to show that the morphisms

$$\Delta_j : Y_j \times_Y X \longrightarrow (Y_j \times_Y X) \times_{Y_j} (Y_j \times_Y X) = Y_j \times_Y X \times_Y X$$

has the corresponding property (i.e., is quasi-compact or a closed immersion). Write $Y_j = \operatorname{colim} Y_{j,\lambda}$ as in Definition 0.1. Replacing $Y_j$ by $Y_{j,\lambda}$ in the formula above, we have the property by our assumption that (3) holds. Since the displayed arrow is the colimit of the arrows $\Delta_{j,\lambda}$ and since we can test whether $\Delta_j$ has the corresponding property by testing after base change by affine schemes mapping into $Y_j \times_Y X \times_Y X$, we conclude by Lemma 0AIA.

Let us prove that (5) implies (1). Let $\{Y_j \to Y\}$ be as in (5). Then we have the fibre product diagram

$$\begin{array}{ccc} \coprod Y_j \times_Y X & \longrightarrow & X \\ \downarrow & & \downarrow \\ \coprod Y_j \times_Y X \times_Y X & \longrightarrow & X \times_Y X \end{array}$$

By assumption the left vertical arrow is quasi-compact or a closed immersion. It follows from Spaces, Lemma 03KD that also the right vertical arrow is quasi-compact or a closed immersion. □